STANDARD OPERATING PROCEDURES (SOP)

“Althea” CRISPR TTP Integrated Bioengineering Architecture

Version 2.0 — Complete Clinical Implementation Protocol

Document Version: 2.0 (EU / ICH GCP R3 Compliant / Final Public Release)

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IN MEMORIAM & ETHICAL DEDICATION

Dedicated to the memory of Althea, whose courageous public journey with Ewing Sarcoma deeply inspired this work. Her legacy of hope and resilience is embedded in the spirit of this protocol.

• The name “Althea” is used symbolically and respectfully to minimize identifying details.

• If future versions or publications include more personal or identifying content, explicit informed consent (or proxy consent from legal guardians/next-of-kin) will be sought, in line with ICMJE standards for publication ethics.

1.0 LEGAL NOTICE & REGULATORY DISCLAIMER

PLEASE READ CAREFULLY BEFORE USING THIS PROTOCOL.

1.1 Nature of the Document

This SOP is a conceptual bioengineering specification, released under CC0 1.0, intended exclusively for use in Phase I/II clinical research planning and scientific discussion.

• Not a marketed product: This architecture is not approved by any regulatory authority (e.g., FDA, EMA, PEI).

• No guarantee of approval: Publication does not imply regulatory authorization.

• Not medical advice: The author is a systems architect, not a physician.

1.2 Regulatory Compliance

• Clinical Trials Regulation (EU) No 536/2014: Trials must be authorized in the Clinical Trials Information System (CTIS) before patient enrollment.

• ICH E6(R3) – Good Clinical Practice: All procedures must comply with the ICH E6(R3) guideline, emphasizing risk-based, quality by design approaches.

• ATMP Regulation (EC 1394/2007): Must apply risk-based development per EMA / CAT guidance.

• German Law (AMG): § 4b, § 13, § 21 apply (PEI approval, GMP license, marketing authorization).

• Warning: Use of this protocol outside a properly authorized clinical trial or legal exemption is illegal and not permitted.

1.3 Pharmacovigilance & Risk Management

• A Risk Management Plan (RMP) per GVP Module V must be established, maintained, and approved by the competent authority.

• Risk Minimisation Measures (RMM) must be designed, approved, and implemented as per GVP Module XVI (Rev 3).

• Evaluation of RMM effectiveness must use methods from GVP XVI Addendum II.

1.4 Transparency & Ethics

• Trials using this SOP must be registered publicly (e.g., EudraCT / CTIS) before enrolling participants.

• All results (positive or negative) should be reported/published.

• Protocol must be approved by an Institutional Review Board (IRB) or Ethics Committee.

• Conduct should follow the Declaration of Helsinki principles.

1.5 Data Protection (GDPR / DSGVO)

• This protocol involves sensitive personal data (e.g., genetic, biometric), covered by Article 9 GDPR.

• A Data Protection Officer (DPO) must review and approve data handling plans, ensuring:

o Mandatory pseudonymisation.

o Split-Key Storage (separation of identity key from analytical data).

o Data minimisation, encryption, and regular DPO audits.

1.6 Limitation of Liability

• This SOP is provided "as is", without warranties.

• All legal, clinical, and regulatory responsibility lies solely with the implementing institution and its licensed staff. This document does not replace necessary licenses, certificates (e.g., GMP), or local SOPs.

2.0 PURPOSE & SCOPE

This SOP describes a clinical research workflow for the “Althea” CRISPR TTP architecture, integrating:

• CRISPR/Cas9 editing of dendritic cells

• GMP-grade dendritic cell production

• Theranostic nanoparticle (TTP) formulation and delivery

• Activation via MR-guided Focused Ultrasound (FUS)

• 3D spatial tumour profiling (e.g., CyClF, spatial transcriptomics)

• AI-driven adaptive dosing and immune monitoring

Intended for Phase I/II research, especially for Ewing Sarcoma (EWSR1-FLI1^{+}).

3.0 PATIENT SCREENING & SELECTION

3.1 Inclusion Criteria

• Histologically confirmed Ewing Sarcoma with EWSR1-FLI1 fusion

• Metastatic disease, including CNS involvement

• Exhausted standard-of-care therapies

• ECOG performance status 0–2

• Adequate organ function

3.2 Exclusion Criteria

• Active autoimmune disease requiring immunosuppression

• Uncontrolled comorbidities

• Pregnancy or breastfeeding

• Known hypersensitivity to any component of the therapy

3.3 Biomarker & Informed Consent Procedures

• Tumour Biopsy: Core-needle biopsy for RNA / NGS analysis

• Informed Consent: Must explicitly address the experimental nature, risks of CRISPR editing, nanoparticle therapy, and data usage protocols.

4.0 CELL PROCESSING (GMP)

4.1 Dendritic Cell Differentiation

Procedure:

Day 0: Thaw PBMCs → adhere 2 h at 37°C, 5% CO_2

Day 1: Add X-VIVO 15 + GM-CSF + IL-4

Day 3: Partial medium change (~50%)

Day 5: Full medium change + replenish cytokines

Day 7: Harvest immature DCs

Quality Control (GMP):

• Grade A/B cleanroom

• Viability > 85%

• Phenotype (CD80 / CD86 / CD83) > 80%

• Endotoxin < 0.5 EU/mL; Mycoplasma negative

4.2 CRISPR Editing of DCs

• Resuspend DCs at 1.2×10^6 cells / 100 μL in Neon Buffer R

• Prepare RNP: 1.8 μM Cas9 + 3.6 μM sgRNA

• Electroporation: 1600 V, 10 ms, 3 pulses

• Recover for 4 h at 37°C, 5% CO_2

• Validate editing by T7 Endonuclease I assay и deep sequencing (NGS) for off-targets

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5.0 NANOPARTICLE (TTP) FORMULATION

5.1 Composition

• Core: Poly(propylene sulfide) + CD99-targeting ligand

• Shell: Hydrogen-bonded organic framework (HOF-101)

• Payload: CRISPR RNP + PD-1 inhibitor + STING agonist

5.2 Quality Control

• Particle size: ~27 ± 3 nm (by DLS), PDI < 0.15

• Loading efficiency: > 85%

• Sterile filtration: 0.22 µm PES membrane

• Endotoxin level: < 0.25 EU/mL

6.0 ADMINISTRATION & ACTIVATION

6.1 Treatment Timeline (21-Day Cycle)

• Week 1: Biopsy → Profiling → DC differentiation → CRISPR editing

• Week 2: Nanoparticle manufacture → QC → IV infusion → FUS activation

• Week 3: Immune monitoring + AI-driven adaptive dosing

6.2 Focused Ultrasound (FUS) Activation

• Device: MR-guided FUS (e.g., ExAblate)

• Frequency: 1–3 MHz; Spatial resolution: 1–2 мм

• Safety: Limit thermal rise to ΔT ≤ 7°C; maintain a margin ≥ 5 мм from critical structures

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7.0 3D SPATIAL PROFILING

• Methods: Cyclic immunofluorescence (CyClF), MIBI, spatial transcriptomics

• Purpose: Map immune microenvironment, guide FUS targeting, assess response

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8.0 SAFETY & MONITORING

8.1 Adverse Event Management

• Cytokine Release Syndrome (CRS): For Grade 3–4, administer Tocilizumab и corticosteroids

• Neurotoxicity: Monitor mental status; escalate with dexamethasone if needed

• Serious Adverse Event (SAE) Reporting: Report SAEs to sponsor и регуляторному органу (например, PEI / BfArM) в течение 24 часов

9.0 AI CO-PILOT PROTOCOL

• Bayesian Optimization: Для схемы дозирования и времени доставки наночастиц

• Federated Learning: Обучение предиктивных моделей между учреждениями с дифференциальной приватностью для защиты данных пациентов

10.0 LEGACY STATEMENT

Honoured Persona: Althea

This architecture, although technically ready, was not implemented clinically due to structural and regulatory barriers. Version 2.0 is published as a memorial blueprint — a call to action for stakeholders to build the frameworks that can deliver for future patients.

11.0 PROTOCOL APPROVAL

Role Name Signature Date

Protocol Author Alexey Mikhailovich Burlai

Quality Assurance

Regulatory Affairs

Principal Investigator

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